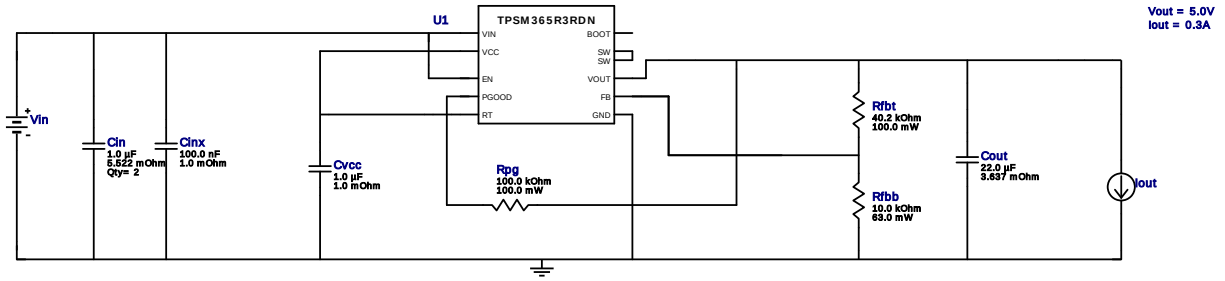
WEBENCH® Design Report

VinMin = 12.0V
VinMax = 24.0V
Vout = 5.0V
Iout = 0.3A

Device = TPSM365R3RDNR
Topology = Buck
Created = 2026-10-12 16:54:51.780
BOM Cost = $2.00
BOM Count = 9
Total Pd = 0.1W

Design : 1 TPSM365R3RDNR
TPSM365R3RDNR 12V-24V to 5.00V @ 0.104A

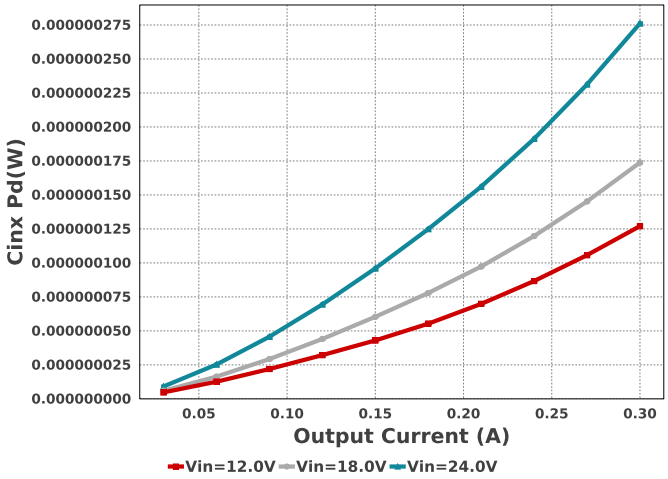


Electrical BOM

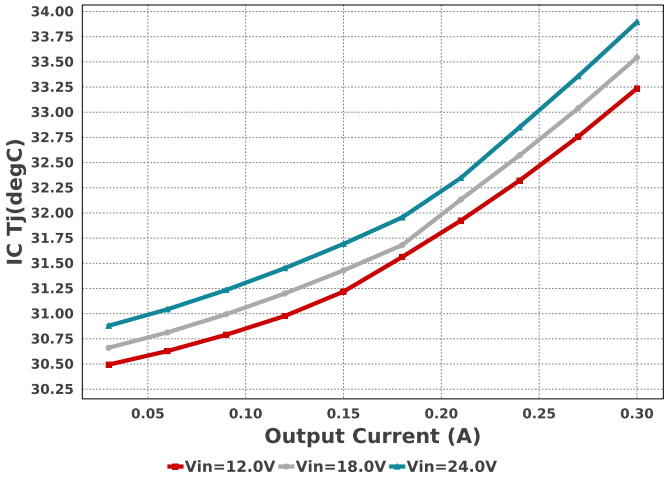
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cin	TDK	CGA3E1X7R1V105K080AC Series= X7R	Cap= 1.0 uF ESR= 5.522 mOhm VDC= 35.0 V IRMS= 2.2162 A	2	$0.05	0603 5 mm^2
Cinx	Yageo	CC0805KRX7R9BB104 Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 0.0 A	1	$0.02	0805 7 mm^2
Cout	Taiyo Yuden	MSASL219LB5226MTNA01 Series= X5R	Cap= 22.0 uF ESR= 3.637 mOhm VDC= 10.0 V IRMS= 3.4179 A	1	$0.07	0805 7 mm^2
Cvcc	Taiyo Yuden	EMK107B7105KA-T Series= X7R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm^2
Rfbb	Vishay-Dale	CRCW040210K0FKED Series= CRCW..e3	Res= 10.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm^2
Rfbt	Yageo	RC0603FR-0740K2L Series= ?	Res= 40.2 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm^2
Rpg	Vishay-Dale	CRCW0603100KFKEA Series= CRCW..e3	Res= 100.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm^2
U1	Texas Instruments	TPSM365R3RDNR	Switcher	1	$1.77	RDN0011A 25 mm^2

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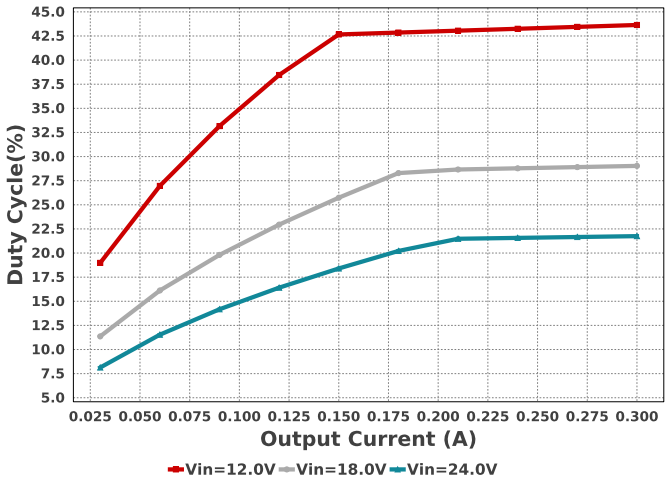
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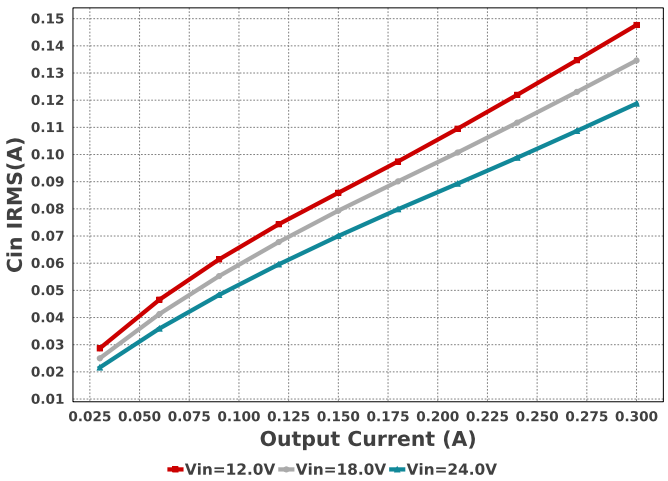
IC Tj



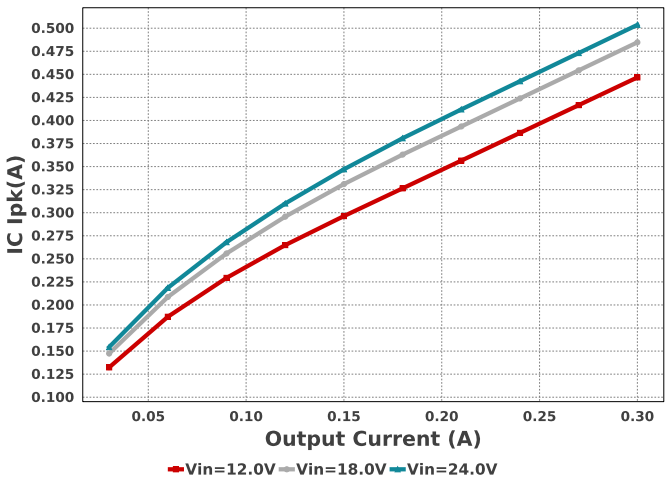
Duty Cycle



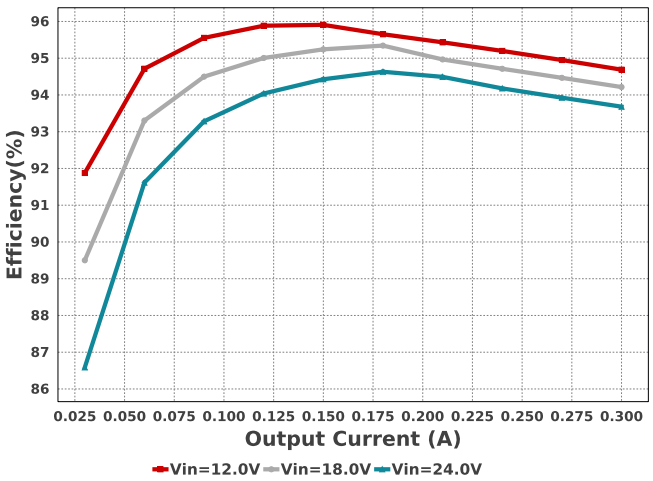
Cin IRMS



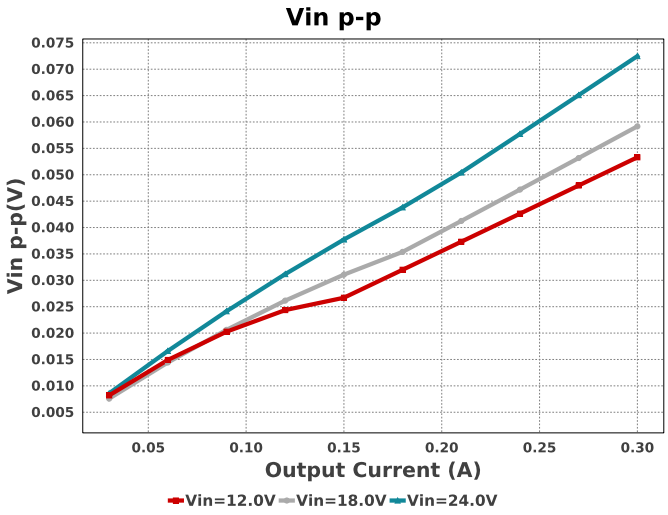
IC Ipk

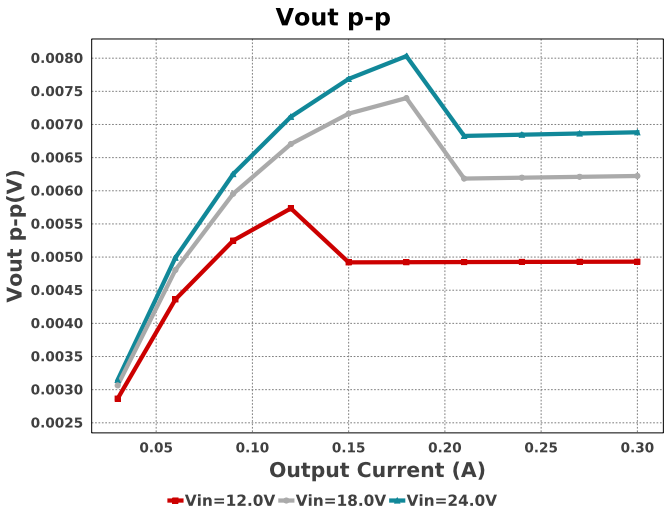


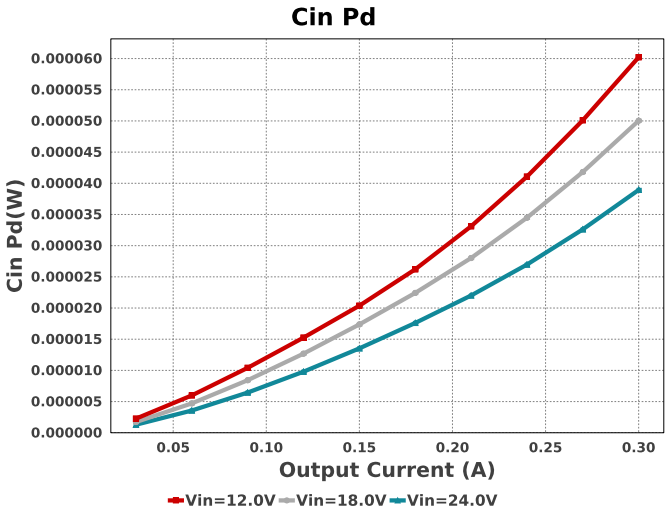
Efficiency

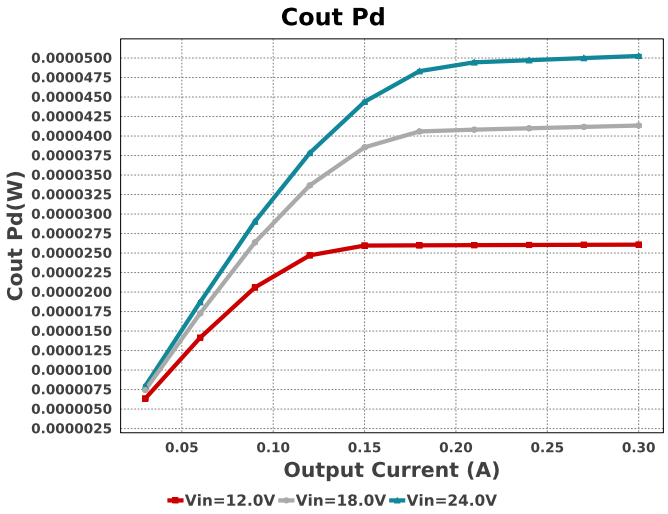


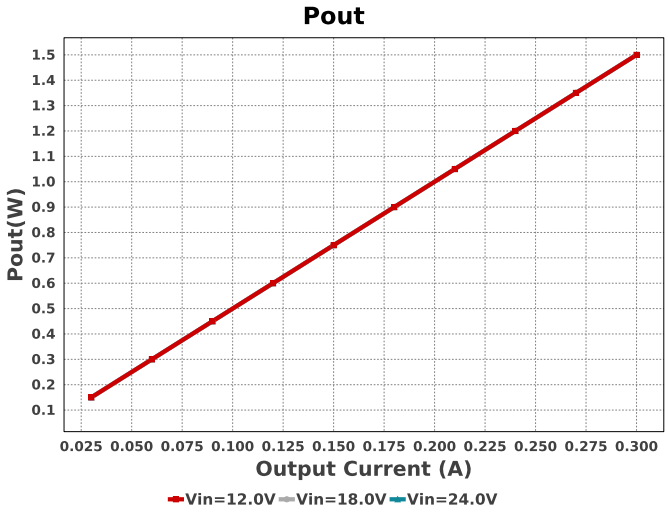
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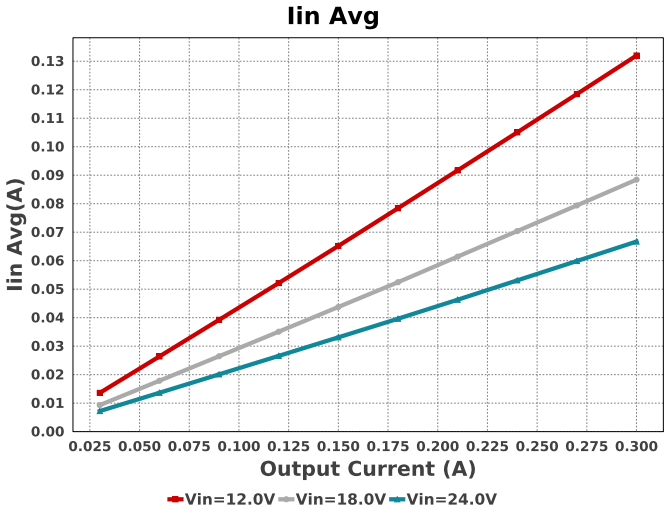




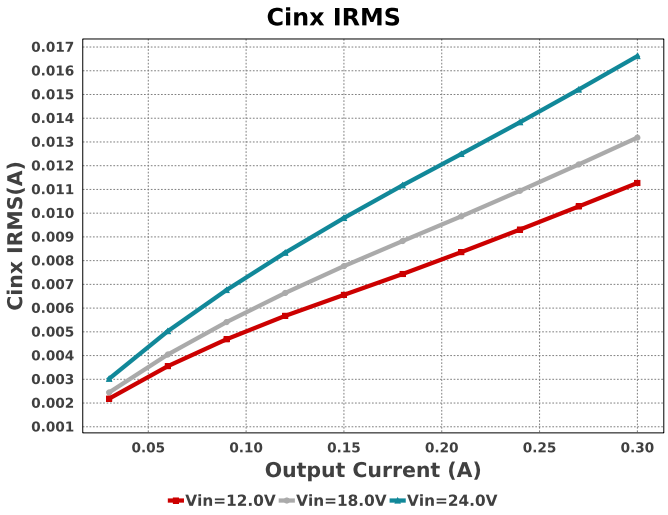


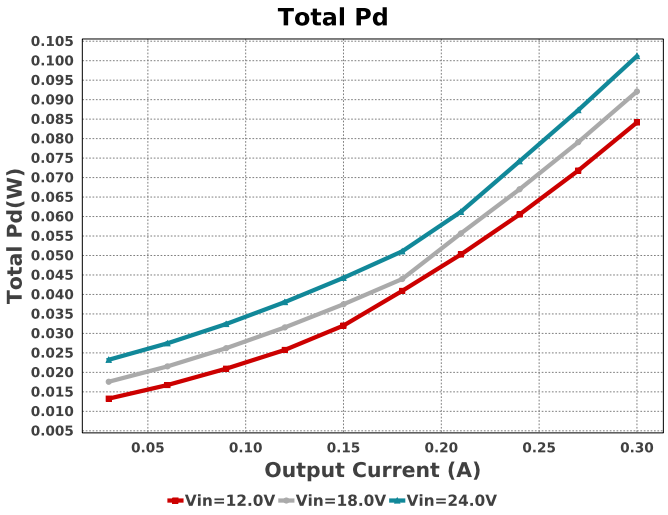


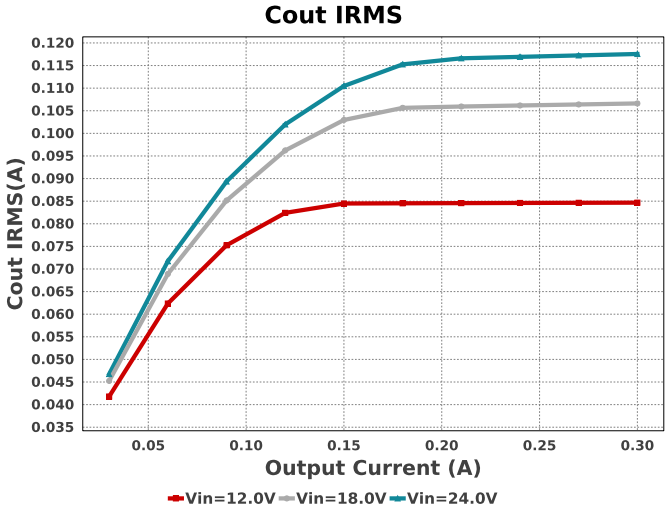


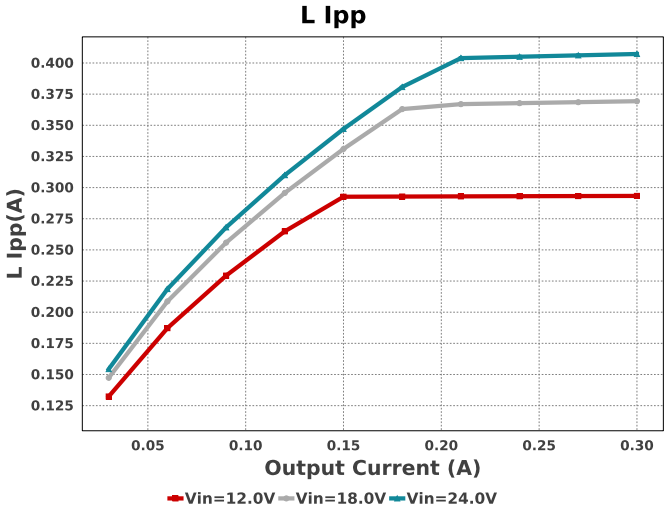


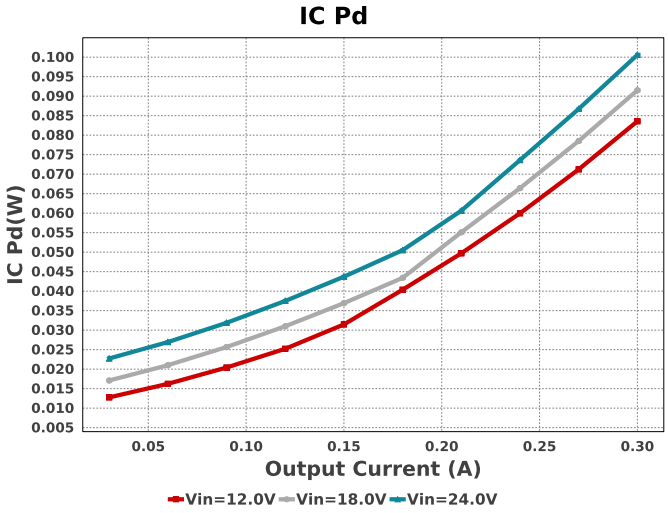
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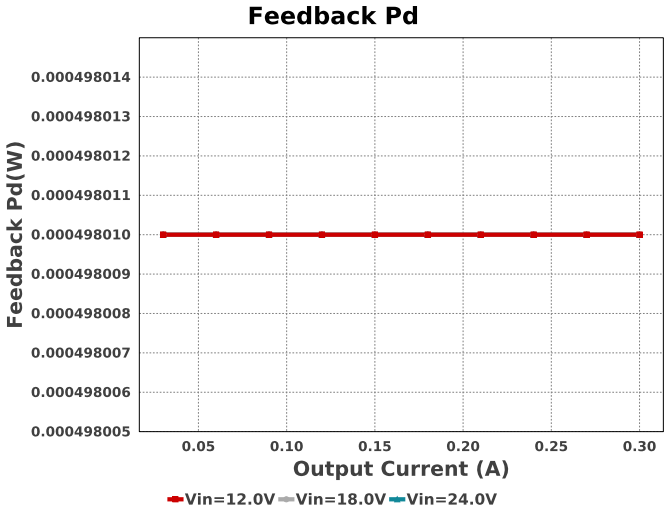




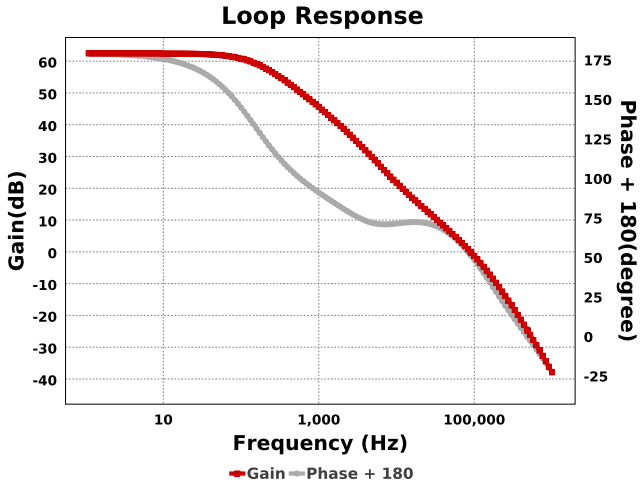


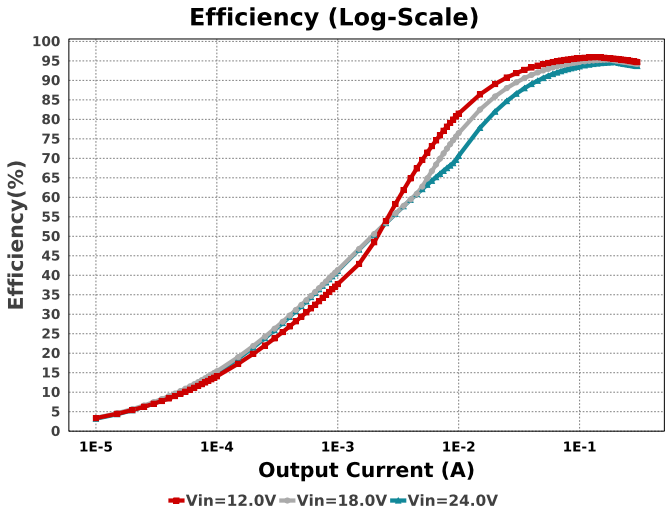






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Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	118.751 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	38.935 µW	Capacitor	Input capacitor power dissipation
3.	Cinx IRMS	16.618 mA	Capacitor	Bulk capacitor RMS ripple current
4.	Cinx Pd	276.15 nW	Capacitor	Bulk capacitor power dissipation
5.	Cout IRMS	117.556 mA	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	50.261 µW	Capacitor	Output capacitor power dissipation
7.	IC Ipk	503.613 mA	IC	Peak switch current in IC
8.	IC Pd	100.59 mW	IC	IC power dissipation
9.	IC Tj	33.895 degC	IC	IC junction temperature
10.	IC Tolerance	25.0 mV	IC	IC Feedback Tolerance
11.	ICThetaJA Effective	38.72 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
12.	Iin Avg	66.716 mA	IC	Average input current
13.	Cin Pd	38.935 µW	Power	Input capacitor power dissipation
14.	Cinx Pd	276.15 nW	Power	Bulk capacitor power dissipation
15.	Cout Pd	50.261 µW	Power	Output capacitor power dissipation
16.	Feedback Pd	498.01 µW	Power	Power Dissipation in Feedback Resistors
17.	IC Pd	100.59 mW	Power	IC power dissipation
18.	Total Pd	101.179 mW	Power	Total Power Dissipation
19.	Feedback Pd	498.01 µW	Resistor	Power Dissipation in Feedback Resistors
20.	BOM Count	9	System Information	Total Design BOM count
21.	Cross Freq	89.506 kHz	System Information	Bode plot crossover frequency
22.	Duty Cycle	21.752 %	System Information	Duty cycle
23.	Efficiency	93.681 %	System Information	Steady state efficiency
24.	FootPrint	65.0 mm^2	System Information	Total Foot Print Area of BOM components
25.	Frequency	1000.0 kHz	System Information	Switching frequency
26.	Gain Marg	-24.786 dB	System Information	Bode Plot Gain Margin
27.	Iout	300.0 mA	System Information	Iout operating point
28.	L Ipp	407.23 mA	System Information	Peak-to-peak inductor ripple current
29.	Low Freq Gain	62.431 dB	System Information	Gain at 1Hz
30.	Mode	CCM	System Information	Conduction Mode
31.	Phase Marg	52.049 deg	System Information	Bode Plot Phase Margin
32.	Pout	1.5 W	System Information	Total output power
33.	Total BOM	$2.0	System Information	Total BOM Cost
34.	Vin	24.0 V	System Information	Vin operating point
35.	Vin p-p	72.482 mV	System Information	Peak-to-peak input voltage

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#	Name	Value	Category	Description
36.	Vout	5.0 V	System Information	Operational Output Voltage
37.	Vout Actual	5.02 V	System Information	Vout Actual calculated based on selected voltage divider resistors
38.	Vout Tolerance	4.158 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
39.	Vout p-p	6.882 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	300.0 m	Maximum Output Current
VinMax	24.0	Maximum input voltage
VinMin	12.0	Minimum input voltage
Vout	5.0	Output Voltage
base_pn	TPSM365R3	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

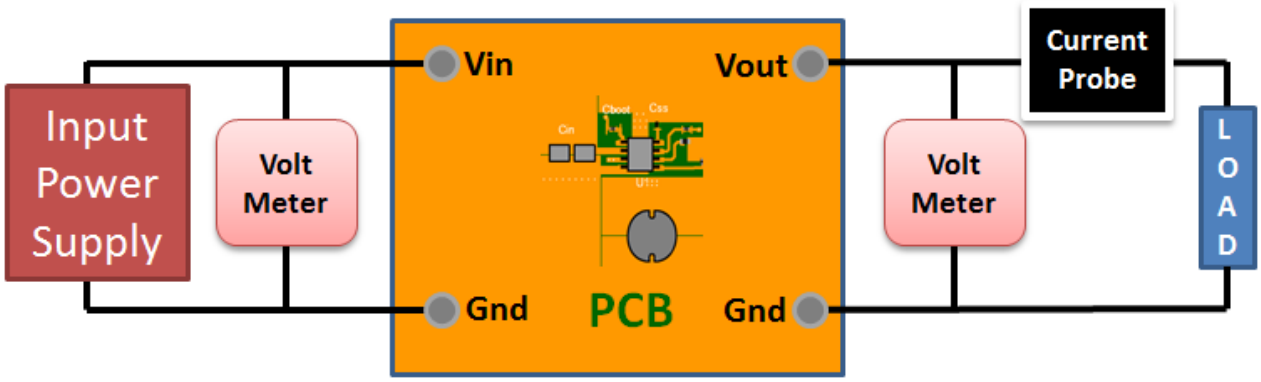
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 12.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : BBF64A2AA795EBD60FCDEAB8F0356F9C[v1]
2. **TPSM365R3** Product Folder : https://www.ti.com/product/TPSM365R3 : contains the data sheet and other resources.

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